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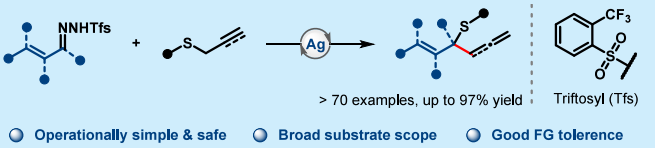
Silver-Catalyzed Doyle−Kirmse Reaction of Allyl Sulfides with Vinyl Triftosylhydrazones

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ABSTRACT: Here, we report a general and practical Doyle−Kirmse reaction of allyl/propargyl sulfides with donor-only vinyl carbenes generated *in situ* from vinyl triftosylhydrazones in the presence of a silver catalyst. This protocol features mild conditions, exhibits a broad substrate scope and exceptional functional group tolerance, and provides corresponding 1,5-dienyl and 1,4-enallenyl sulfides in high yields. Moreover, gram-scale synthesis, late-stage modifications of complex molecules, and post-synthetic transformations were performed to demonstrate the applicability of this protocol.



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[2,3]-Sigmatropic rearrangements of onium ylides are an important and fundamental transformation in organic chemistry for synthesizing valuable heteroatom-containing compounds with high stereoselectivity.[1] Among these, the [2,3]-sigmatropic rearrangement of sulfonium ylides generated from carbenes and allyl/propargyl sulfides, known as the Doyle−Kirmse (D−K) reaction,[2] offers a potentially general method for the stereoselective construction of a high-value sulfur-containing quaternary or tertiary carbon center, which has found applications in the synthesis of bioactive molecules and natural products. Over the decades, various transition metal and metal-free D−K reactions using tetramethylsilane (TMS)-stabilized[3] or electron-withdrawing group (EWG)-derived diazo reagents[4−7] as well as a few donor-only diazomethane reagents[8] and *N*-sulfonyl-1,2,3-triazoles[9] as carbene precursors have been established. In contrast, the utilization of α-vinyl diazo compounds in D−K reactions remains challenging and scarcely explored because of their propensity to undergo intramolecular cyclization to form pyrazoles.[10] Despite these challenges, Wang and Feng's groups successfully employed vinyl diazoacetates[4e] or vinyldiazo compounds with chiral Oppolzer's camphor sultam[6a] or pyrazoleamide[6b] auxiliary as vinylcarbene precursors for the enantioselective D−K reaction (Figure 1A). However, to the best of our knowledge, there has been no general catalytic system for the D−K reaction with donor-substituted α-vinyldiazomethanes as carbene precursors to date due to their inherent reactivity. A single example of a rhodium-catalyzed D−K reaction of phenyl allyl sulfide with *in situ* generated vinyldiazomethane from tosylhydrazone was found in a report by Wang and co-workers, although providing the desired product in 48% yield.[8a] In 2015, Wang and co-workers used cyclopropenes as vinylcarbene precursors in the D−K reaction to bypass the general reactivity issues of α-vinyldiazomethanes as donor diazo compounds (Figure 1A).[11]

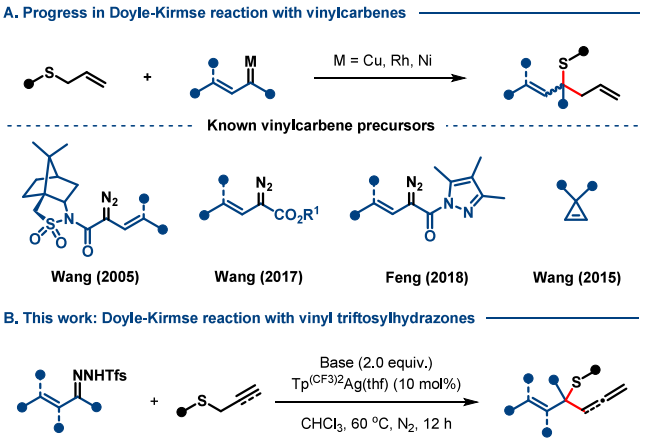


Figure 1. Background and motivation of the present work.

Recently, we discovered vinyl triftosylhydrazones as easily decomposable diazo surrogates and addressed the general issue of the reactivity of α-vinyldiazomethane toward pyrazole formation by merging it with a bulky silver catalyst to generate highly electrophilic silver vinylcarbenes.[12] Encouraged by our success in transition-metal-catalyzed carbene transfer reactions using vinyl triftosylhydrazones as vinylcarbene precursors,[13,14] we envision that the merger of vinyl triftosylhydrazones with silver catalysis might govern the highly efficient D−K reaction

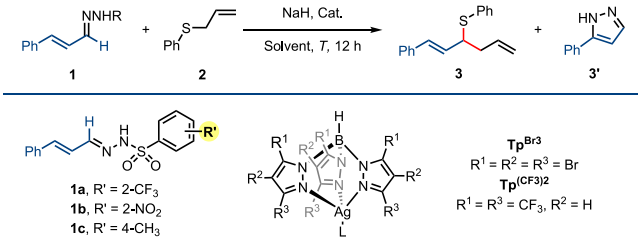
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Table 1. Optimization of the Reaction Conditions[a]



entry	1	catalyst	solvent	T (°C)	3 (%)[b]	3′ (%)[b]
1	**1a**	$Tp^{Br_3}Ag(thf)$	$CHCl_3$	60	82	trace
2	**1a**	$Tp^{(CF_3)_2}Ag(thf)$	$CHCl_3$	60	97 (93)	trace
3	**1a**	AgOTf	$CHCl_3$	60	60	23
4	**1a**	$Cu(CH_3CN)_4PF_6$	$CHCl_3$	60	28	59
5[c]	**1a**	$Rh_2(OAc)_4$	$CHCl_3$	60	67	11
6	**1a**	$Tp^{(CF_3)_2}Ag(thf)$	DCM	60	55	16
7	**1a**	$Tp^{(CF_3)_2}Ag(thf)$	$PhCF_3$	60	78	trace
8	**1a**	$Tp^{(CF_3)_2}Ag(thf)$	1,4-dioxane	60	trace	53
9	**1a**	$Tp^{(CF_3)_2}Ag(thf)$	$CHCl_3$	80	85	trace
10	**1a**	$Tp^{(CF_3)_2}Ag(thf)$	$CHCl_3$	40	41	trace
11	**1b**	$Tp^{(CF_3)_2}Ag(thf)$	$CHCl_3$	60	76	8
12	**1c**	$Tp^{(CF_3)_2}Ag(thf)$	$CHCl_3$	60	64	13

[a]Reaction conditions: compound **1** (0.3 mmol), allyl phenyl sulfide **2** (0.6 mmol), NaH (0.6 mmol), and catalyst (10 mol %) in $CHCl_3$ (6 mL) at the specified temperature for 12 h. [b]The yield was determined by ^1H NMR using CH_2Br_2 as the internal standard. [c]With 1 mol % of catalyst. The yield in parentheses refers to the isolated yield.

on α-vinyldiazomethanes. Herein, we report a general and efficient silver-catalyzed D−K reaction between allyl/propargyl sulfides and α-vinyldiazomethanes derived from vinyl triftosylhydrazones (Figure 1B). This approach provides practical access to diverse 1,5-dienyl or 1,4-enallenyl sulfides with high efficiency. Dienyl sulfides are valuable core structures in biologically active molecules and versatile building blocks for the synthesis of value-added products.[15]

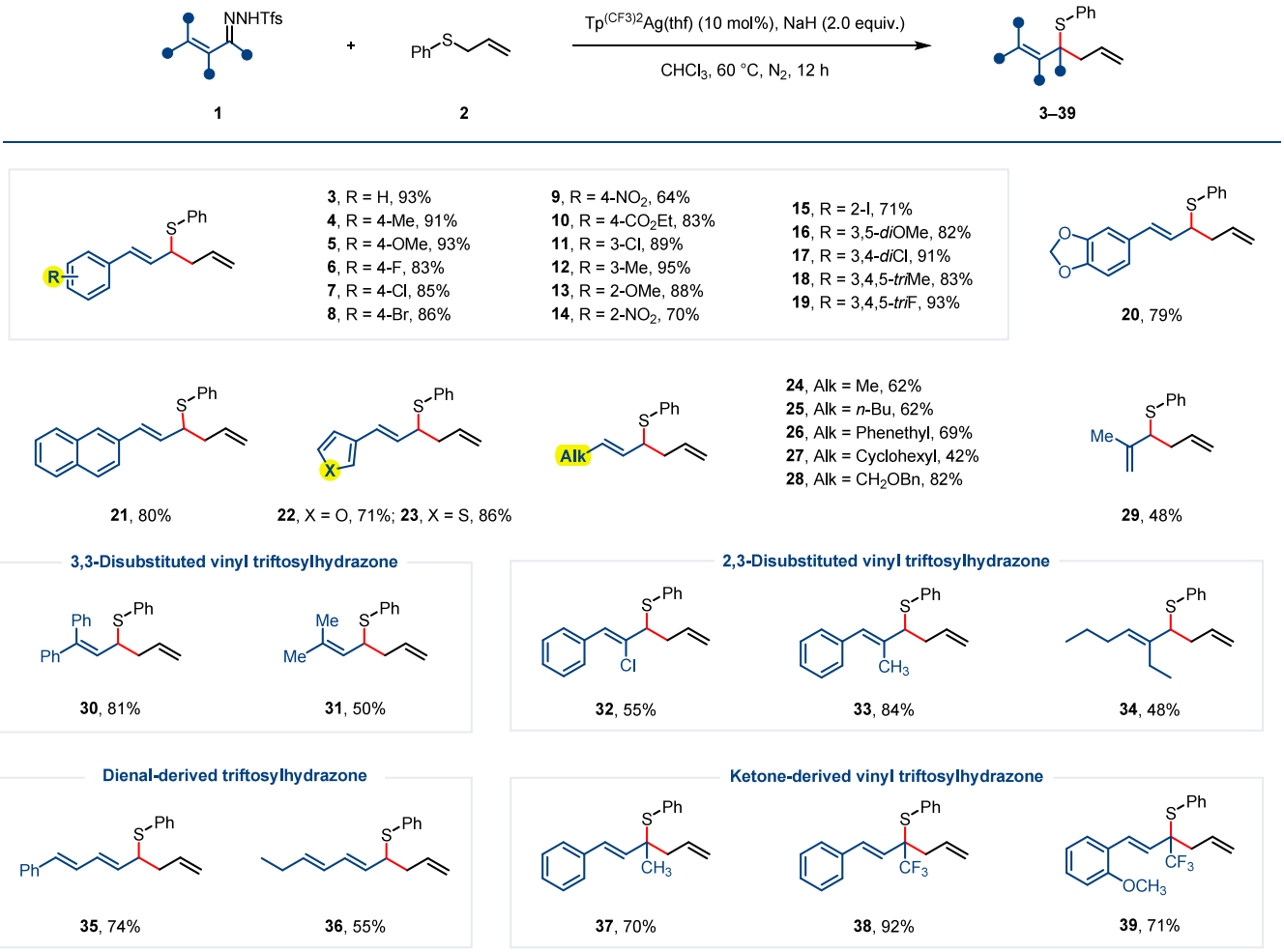
To test our hypothesis, we employed the reaction of α-cinnamyl *N*-sulfonyl hydrazone **1** and allyl phenyl sulfide **2**, using NaH as a base, as a model system to identify a suitable catalyst (Table 1; for details, see Tables S1−S5 of the Supporting Information). Initially, the reaction using bulky $Tp^{Br_3}Ag(thf)$ as a catalyst in chloroform ($CHCl_3$) at 60 °C provided the desired product **3** in 82% nuclear magnetic resonance (NMR) yield (entry 1 in Table 1). Delightfully, the relatively more electrophilic silver catalyst $Tp^{(CF_3)_2}Ag(thf)$ produced the target product **3** in superior yield (97% NMR yield, entry 2). In both cases, only a negligible amount of pyrazole **3′** was detected. Other catalysts commonly used for D−K reactions, such as AgOTf, $Cu(CH_3CN)_4PF_6$, and $Rh_2(OAc)_4$,[4c,e,6a,8] afforded the desired product **3**, albeit in a decreased yield along with more undesired product **3′** (entries 3−5). Other solvents, such as dichloromethane (CH_2Cl_2) and trifluorotoluene ($PhCF_3$), did not improve the yield of product **3**, but 1,4-dioxane yielded pyrazole (**3′**) as a major product (entries 6−8). Increasing or decreasing the reaction temperature could negatively affect the reaction efficiency (entries 9 and 10). The reactions involving *N*-nosylhydrazone **1b**[16] and *N*-tosylhydrazone **1c**[17] also proceeded well but afforded the desired product **3** in lower yields (entries 11 and 12), suggesting the superior performance of triftosylhydrazone compared to other *N*-sulfonyl hydrazones.[18] Interestingly, when an excessive amount of triftosylhydrazone **1a** was employed, product **3** preferentially underwent cyclopropylation instead of proceeding with the second D−K rearrangement (Table S5 of the Supporting Information).

With the optimized reaction conditions in hand, we explored the scope of vinyl triftosylhydrazones derived from various α,β-unsaturated aldehydes, using allyl phenyl sulfide **2** as the reaction partner (Scheme 1). A variety of α-(arylvinyl) triftosylhydrazones, bearing electron-withdrawing or electron-donating substituents at the *para*, *meta*, or *ortho* position of the aryl ring, were well-tolerated and provided the desired 1,5-dienyl sulfides **3**−**20** in high to excellent yields. Vinyl triftosylhydrazones containing medicinally relevant (hetero)-aryl rings, such as naphthalene, furan, and thiophene, underwent the reaction smoothly, affording the target products **21**−**23** in good yields. This silver-catalyzed system was also applicable to vinyl triftosylhydrazones bearing non-aryl substituents, including methyl (**24** and **29**), *n*-butyl (**25**), phenylethyl (**26**), cyclohexyl (**27**), and (phenylmethoxy)-methyl (**28**) groups. Likewise, a set of 3,3-disubstituted and 2,3-disubstituted vinyl triftosylhydrazones could also be effectively activated by the current conditions, leading to the formation of multisubstituted 1,5-dienyl sulfide products **30**−**34** in 48−84% yields. 2,4-Pentadienal-derived triftosylhydrazones also serve as suitable substrates, furnishing corresponding 1,5,7-trienyl sulfide products **35** and **36** in 74 and 55%, respectively, which are common skeletons in bioactive natural products and drug molecules.[19] Finally, we examined the reactivity of vinyl triftosylhydrazones derived from α,β-unsaturated ketones, which underwent smooth reaction to deliver desired 1,5-dienyl sulfides containing a quaternary carbon center in good yields (**37**−**39**). These compounds represent the most potent inhibitors of stromelysin-1.[20]

Next, we surveyed the scope of allyl sulfides using vinyl triftosylhydrazone **1a** as the constant coupling reagent (Scheme 2A). Aryl allyl sulfides substituted with various substituents, such as methyl, methoxy, isopropyl, halogens, and trifluoromethyl groups, reacted smoothly to afford correspond-

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Scheme 1. Scope of Vinyl Triftosylhydrazones[a]



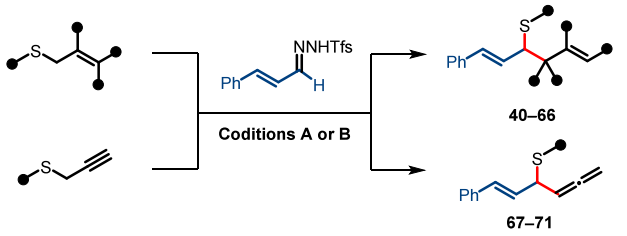
[a]Standard conditions: compound **1** (0.3 mmol), allyl phenyl sulfide **2** (0.6 mmol), NaH (2.0 equiv), $Tp^{(CF_3)_2}Ag(thf)$ (10 mol %), and $CHCl_3$ (6 mL), under N_2 for 12 h. In all cases, isolated yields are shown.

ing products **40**−**51** in good to excellent yields. Replacing the phenyl ring of aryl allyl sulfides with electron-rich naphthalene (**52**) and thiophene (**53**) rings afforded the desired product in high yields, while 2-methyl furyl allyl sulfide (**54**) resulted in a poor yield. Similarly, benzyl-, alkyl-, alkyl ester-, and allyl-substituted allyl sulfides (**55**−**59**) were also compatible with the reaction conditions. In the case of diallyl sulfide, the desired product **59** was exclusively obtained in 80% yield under standard conditions. Intriguingly, upon adjustment of the stoichiometric ratio of compound **1a** to diallyl sulfide from 1:2 to 2:1, no detectable double D−K reaction product was observed. Instead, product **59** was isolated in 63% yield, along with side products resulting from cyclopropanation and self-cyclization (Scheme S2 of the Supporting Information). The reaction demonstrated a high tolerance of various functionalities, such as methyl (**60**), phenyl (**61**), halogens (**62** and **63**), and azide (**64**), on the vinyl unit of aryl allyl sulfides, furnishing target products in 51−85% yields. Sterically hindered 1,1-dimethylallyl aryl sulfide could still provide the desired product **65** in a high yield. Likewise, the current protocol was suitable for cyclohex-2-enyl aryl sulfide, giving desired product **66** in 49% yield with moderate diastereoselectivity (49% yield, dr = 7:2). Finally, the scope of this reaction was successfully extended to propargyl sulfides, thereby providing a convenient route to synthetically useful 1,4-enallenyl sulfides (Scheme 2B).[21] However, a slight modification from the original conditions, such as changing the base from NaH to CsF, is necessary to achieve a synthetically useful yield (see Tables S6−S9 of the Supporting Information). Under these modified conditions, aryl propargyl sulfides with methyl, isopropyl, *tert*-butyl, and chloro functionality at the *para* position of the aryl ring reacted smoothly to afford corresponding 1,4-enallenyl sulfides **67**−**71** in 50−58% yields.

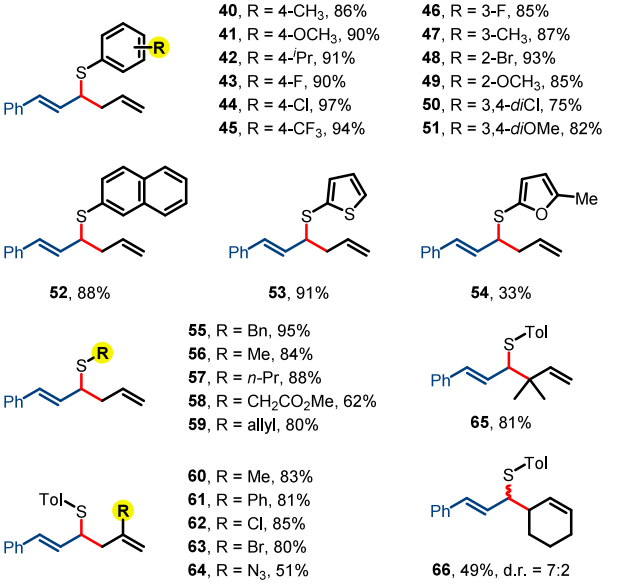
To further demonstrate the practicality of this protocol, we performed a 5 mmol scale reaction between vinyl triftosylhydrazone **1a** and allyl phenyl sulfide **2**, resulting in the successful formation of product **3** in 92% (1.23 g) isolated yield (Scheme 3A). The synthetic value of formed products **3** and **64** was then demonstrated. Oxidation of compound **3** with *m*-chloroperbenzoic acid (*m*-CPBA) produced sulfone **72** in 90% yield. Pleasingly, compound **64** could undergo Cu(I)-catalyzed azide−alkyne cycloaddition to obtain medicinally significant triazole **74** in 95% yield.[22] The late-stage modification of complex molecules, such as perillaldehyde and fluvastatin-decorated vinyl triftosylhydrazones, was also possible, giving products **75** and **76** in 54 and 71% yields,

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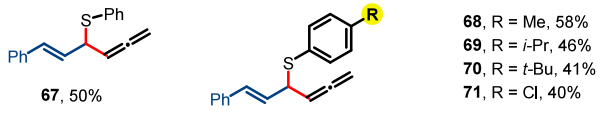
Scheme 2. Scope of Allyl and Propargyl Sulfides[a,b]



A. Scope of allyl sulfides (Conditions A)



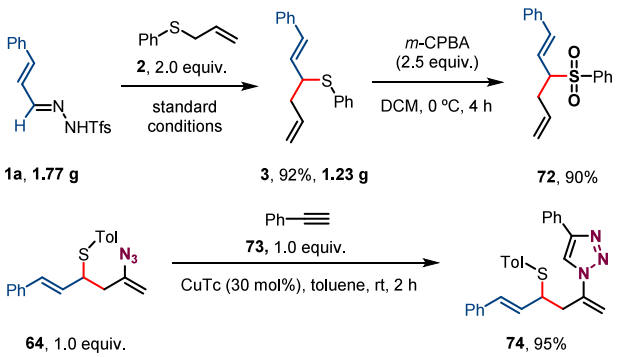
B. Scope of propargyl sulfides (Conditions B)



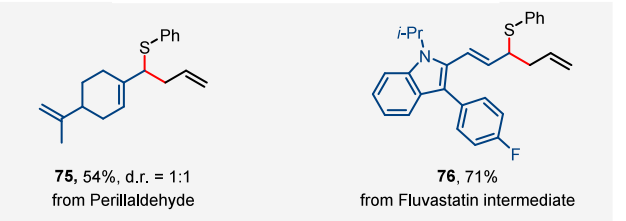
[a]Condition A: vinyl triftosylhydrazone **1a** (0.3 mmol), allyl sulfides **2** (0.6 mmol), NaH (2.0 equiv), $Tp^{(CF_3)_2}Ag(thf)$ (10 mol %), and $CHCl_3$ (6 mL), under N_2 for 12 h. [b]Condition B: vinyl triftosylhydrazones **1** (0.3 mmol), propargyl sulfides (0.9 mmol), CsF (2.0 equiv), $Tp^{(CF_3)_2}Ag(thf)$ (10 mol %), and $CHCl_3$ (6 mL), under N_2 for 12 h. Tol = *p*-methyl phenyl.

Scheme 3. Scale-Up Synthesis, Synthetic Applications, and Control Experiments

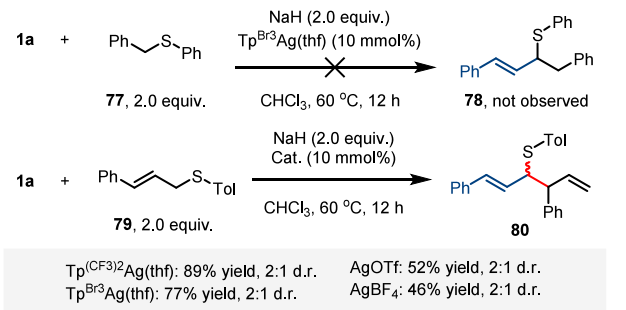
A. Gram-scale synthesis and synthetic applications



B. Late-stage modification of natural products



C. Control experiments



respectively (Scheme 3B), highlighting their potential in medicinal chemistry.

Finally, a series of control experiments were conducted to elucidate the reaction mechanism (Scheme 3C). First, the reaction of vinyl triftosylhydrazone **1a** with benzyl phenyl sulfide **77** failed to yield the Stevens rearrangement product **78** under standard conditions, which could rule out a probable [1,2]-sigmatropic rearrangement. Next, we performed reactions between vinyl triftosylhydrazone **1a** and cinnamyl tolyl sulfide **79** with different Ag(I) catalysts. The diastereoselectivity (dr) of product **80** was consistently determined to be 2:1, regardless of the silver catalyst or its associated ligands. Based on the above observations and previous literature, this sivler-catalyzed [2,3]-sigmatropic rearrangement likely proceeds via a metal-free ylide intermediate (Scheme S4 of the Supporting Information).[4c,e,g,8b]

In conclusion, we have demonstrated that the merger of vinyl triftosylhydrazones with silver catalysis enabled the highly efficient D−K reaction of sulfides with donor-substituted vinyldiazomethanes. The bulky and weakly coordinated $Tp^{(CF_3)_2}Ag$ catalyst is capable of activating donor-only *α*-vinyldiazomethanes under mild conditions toward the D−K reaction of diverse allyl/propargyl sulfides, affording medicinally interesting 1,5-dienyl and 1,4-enallenyl sulfides with high catalytic efficiency. The asymmetric D−K reaction of allyl sulfides with vinyl triftosylhydrazones is under exploration and will be reported in due course (for preliminary experimental results, see Scheme S1 of the Supporting Information).

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c01059.

Experimental details, characterization data, and NMR spectra copies (PDF)

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Notes

The authors declare no competing financial interest.

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